Python Programming - 2301CS404

Lab - 6

Roll No. : 324

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01) WAP to find sum of all the elements in a List.

In [1]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

ans = 0;

for i in l1:
    ans = ans + i
print(ans)
```

```
55
```

02) WAP to find largest element in a List.

In [2]:
```
l1 = list(map(int,input("Enter Data using space:").split()))
maxEle = l1[0]
```

```
for i in l1:
    if i>maxEle:
        maxEle = i;
print(maxEle)
```

```
10
```

03) WAP to find the length of a List.

In [3]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

print(len(l1))
```

```
10
```

04) WAP to interchange first and last elements in a list.

In [4]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

temp = l1[0]
l1[0] = l1[-1]
l1[-1] = temp

print(l1)
```

```
[10, 2, 3, 4, 5, 6, 7, 8, 9, 1]
```

05) WAP to split the List into two parts and append the first part to the end.

In [9]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

newList = l1[len(l1)//2::]
newList.extend(l1[0:len(l1)//2:])

print(newList)
```

```
[6, 7, 8, 9, 10, 1, 2, 3, 4, 5]
```

06) WAP to interchange the elements on two positions entered by a user.

In [1]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

pos1 = int(input("Enter First Position"))
pos2 = int(input("Enter Second Position"))

temp = l1[pos1]
l1[pos1] = l1[pos2]
l1[pos2] = temp

print(l1)
```

```
[1, 2, 3, 8, 5, 6, 7, 4, 9, 10]
```

07) WAP to reverse the list entered by user.

In [4]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

print(l1[-1::-1])
```

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

08) WAP to print even numbers in a list.

In [5]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

for i in l1:
    if i%2==0:
        print(i,end=" ")
```

```
2 4 6 8 10
```

09) WAP to count unique items in a list.

In [9]:
```
l1 = input("Enter Data using space:").split()
count = 0
for i in l1:
    if l1.count(i)==1:
        count = count + 1

print(count)
```

```
3
```

10) WAP to copy a list.

In [13]:
```
l1 = input("Enter Data using space:").split()

l2 = l1.copy()

print(l2)
```

```
['1', '2', '3', '4', '5']
```

11) WAP to print all odd numbers in a given range.

In [7]:
```
num1 = int(input("Enter First Number:"))
num2 = int(input("Enter Second Number:"))

l1 = []

for i in range(num1,num2+1):
    if i%2!=0:
        l1.append(i)
```

```
print(l1)
```

```
[11, 13, 15, 17, 19]
```

12) WAP to count occurrences of an element in a list.

In [8]:
```
l1 = list(map(int,input("Enter Data using space:").split()))
idx = int(input("Enter element:"))

print(l1.count(idx))
```

```
3
```

13) WAP to find second largest number in a list.

In [16]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

l1.sort()

# secondLargest = l[-1]

for i in l1[-2::-1]:
    if i< l1[-1]:
        print(i)
        break;
```

```
3
```

14) WAP to extract elements with frequency greater than K.

In [19]:
```
l1 = input("Enter Data using space:").split()

ans = []

k = int(input("Enter K:"))

for i in l1:
    if l1.count(i) > k and i not in ans:
        ans.append(i)
print(ans)
```

```
['1', '2']
```

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

In [20]:
```
l1 = []

for i in range(0,10):
    l1.append(i**2)
print(l1)
```

```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

In [23]:
```
l1 = input("Enter Data using space:").split()

ans = []

for i in l1:
    if i[0] == 'b':
        ans.append(i)
print(ans)
```

```
['banana', 'babaco', 'berry']
```

17) WAP to create a list of common elements from given two lists.

In [21]:
```
l1 = input("Enter Data using space:").split()
l2 = input("Enter Data using space:").split()

ans = []

for i in l1:
    if i in l2 and i not in ans:
        ans.append(i)
print(ans)
```

```
['1', '2', '3', '4']
```

In [ ]: